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Program: BTech CSE (B16)

HDOC Day-14 – C Programming

Problem Statement

Write a **menu-driven C program using switch-case** to input a positive integer and perform one of the following operations:

1. Check whether **all digits are even**.
2. Check whether **all digits are prime**.

For a single digit, the prime digits are only **2, 3, 5, and 7**.

1. Algorithm

Step 1

Start.

Step 2

Declare integer variables:

`num, temp, digit, choice, flag`

Step 3

Input a positive integer `num`.

Step 4

Display the menu:

1. Check whether all digits are even
2. Check whether all digits are prime

Step 5

Input the user's choice.

Step 6

Use `switch(choice)` to select the required operation.

Case 1 — Check Whether All Digits Are Even

1. Set `temp = num`.
2. Set `flag = 1`, assuming all digits are even.
3. Extract the last digit:

```
digit = temp % 10
```

4. Check:

```
digit % 2 != 0
```

5. If true, an odd digit is found:
 - o Set `flag = 0`.
 - o Exit the loop using `break`.
6. Otherwise, remove the last digit:

```
temp = temp / 10
```

7. Continue until all digits are checked.
8. If `flag == 1`, display "**All digits are even**".
9. Otherwise, display "**All digits are not even**".

Case 2 — Check Whether All Digits Are Prime

1. Set `temp = num`.
2. Set `flag = 1`, assuming all digits are prime.
3. Extract each digit:

```
digit = temp % 10
```

4. Check whether the digit is one of the prime digits:

```
2, 3, 5, 7
```

5. If the digit is not 2, 3, 5, or 7:
 - o Set flag = 0.
 - o Exit the loop.
6. Otherwise, remove the digit using:


```
temp = temp / 10
```
7. Continue until all digits are checked.
8. If flag == 1, display "**All digits are prime**".
9. Otherwise, display "**All digits are not prime**".

Step 7

For any other menu choice, display "**Invalid choice**".

Step 8

Stop.

2. Test Case Table

Test Case Number	Choice	Expected Output	Actual Output	Result
1	12345 1	All digits are not even	All digits are not even	Pass
2	24680 1	All digits are even	All digits are even	Pass
3	4826 1	All digits are even	All digits are even	Pass
4	2357 2	All digits are prime	All digits are prime	Pass
5	2379 2	All digits are not prime	All digits are not prime	Pass
6	25372 2	All digits are prime	All digits are prime	Pass

3. C Program

```
#include <stdio.h>

int main()
{
    int num, temp, digit, choice, flag;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\nMENU\n"); CC
```

```

printf("1. Check whether all digits are even\n");
printf("2. Check whether all digits are prime\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        temp = num;
        flag = 1;

        while(temp > 0)
        {
            digit = temp % 10;

            if(digit % 2 != 0)
            {
                flag = 0;
                break;
            }

            temp = temp / 10;
        }

        if(flag == 1)
            printf("All digits are even\n");
        else
            printf("All digits are not even\n");

        break;

    case 2:
        temp = num;
        flag = 1;

        while(temp > 0)
        {
            digit = temp % 10;

            if(digit != 2 && digit != 3 &&
               digit != 5 && digit != 7)
            {
                flag = 0;
                break;
            }

            temp = temp / 10;
        }

        if(flag == 1)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");

        break;
}

```

```
        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

4. Compile and Execute

Save the program as:

day14.c

Compile using the **cc compiler**:

```
cc day14.c
```

Execute:

```
./day14
```

```
(kali㉿kali)-[~/Desktop]
$ cc day14.c

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 24680

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 4826

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 2357

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 2379

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 25372

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
```

5. Testing Against Prepared Use Cases

Test Case 1 — All Digits Even

Input:

Enter a positive integer: 24680

MENU

1. Check whether all digits are even
 2. Check whether all digits are prime
- Enter your choice: 1

Digits checked:

2 → Even
4 → Even
6 → Even
8 → Even
0 → Even

Expected Output:

All digits are even

Actual Output: All digits are even

Result: PASS ✓

Test Case 2 — All Digits Not Even**Input:**

Enter a positive integer: 12345
Enter your choice: 1

Since digits 1, 3, 5 are odd, all digits are not even.

Expected Output:

All digits are not even

Actual Output: All digits are not even

Result: PASS ✓

Test Case 3 — All Digits Prime**Input:**

```
Enter a positive integer: 2357
Enter your choice: 2
```

Digits:

```
2 → Prime
3 → Prime
5 → Prime
7 → Prime
```

Therefore, every digit is prime.

Expected Output:

```
All digits are prime
```

Actual Output: All digits are prime

Result: PASS ✓

Test Case 4 — All Digits Not Prime

Input:

```
Enter a positive integer: 2379
Enter your choice: 2
```

Digits 2, 3, and 7 are prime, but **9 is not prime**.

Expected Output:

```
All digits are not prime
```

Actual Output: All digits are not prime

Result: PASS ✓

Final Result

The menu-driven C program was successfully **written, compiled, executed, and tested**. The program correctly uses **switch-case**, loops, conditional statements, and a `flag` variable to determine whether all digits are even or all digits are prime. All prepared test cases **PASS ✓**